Unit: 6 Unit 5: Linear Equations
Topic: Parallel Lines

Name: ____________
Date: ________

Advanced (Habanero)

Q1. Which of the following lines are parallel to the line $y = 2x + 1$?
(A) $y = 2x - 1$
(B) $y = 3x + 1$
(C) $y = x + 2$
(D) $y = -2x + 1$
(E) $y = 2x + 2$

Q2. If two lines have the same y-intercept and are parallel, what can be said about their slopes?
(A) They are equal
(B) They are opposite
(C) They are perpendicular
(D) They are equal in magnitude but opposite in sign
(E) They are not related

Q3. What is the equation of a line parallel to $y = 3x - 2$ that passes through $(1, 4)$?
(A) $y = 3x + 1$
(B) $y = 3x - 1$
(C) $y = 2x + 2$
(D) $y = x + 3$
(E) $y = 3x + 2$

Q4. Which of the following is a characteristic of parallel lines?
(A) They intersect at a right angle
(B) They have the same slope
(C) They have the same y-intercept
(D) They are perpendicular to each other
(E) They never intersect

Q5. If $y = mx + b$ and $y = nx + c$ are parallel lines, what is the relationship between m and n?
(A) $m = n$
(B) $m = -n$
(C) $m =$
rac1n
(C) $m = -$
rac1n
(C) $meqn$

Q6. What is the slope of a line parallel to $y = -4x + 2$?
(A) -4
(B) 4
(C)
rac14
(C) $-$
rac14
(C) 2

Q7. Two lines have equations $y = 2x + 1$ and $y = 2x - 3$. What can be said about these lines?
(A) They are parallel
(B) They are perpendicular
(C) They intersect at $(0, 1)$
(D) They are skew
(E) They are the same line

Q8. If a line has a slope of -2 and y-intercept of 3, what is the equation of a line parallel to it?
(A) $y = -2x + 3$
(B) $y = -2x - 3$
(C) $y = 2x + 3$
(D) $y = -2x + 1$
(E) $y = 2x - 3$

Open Ended Questions (FRQ)

Q9. Prove that if two lines have the same slope, they are parallel. Show all work and explain your reasoning.

Q10. Find the equation of a line parallel to $y = x - 2$ that passes through the point $(2, 3)$. Show all work and explain your reasoning.

Chef's Tips

- Ensure students understand the concept of parallel lines and their characteristics
- Encourage students to show all work and explain their reasoning for free-response questions
- Remind students to use the slope-intercept form of a line ($y = mx + b$) to find equations of parallel lines